Supplementary Information

An Integrated TOPSIS Framework with Full-Range Weight Sensitivity Analysis for Robust Decision Analysis

Àlex Gaona, Albert Guisasola and Juan Antonio Baeza.

GENOCOV, Departament d'Enginyeria Química Biològica i Ambiental. Escola d'Enginyeria. Universitat Autònoma de Barcelona, 08193, Bellaterra (Barcelona), Spain.

Email: Alex.Gaona@uab.cat, Albert.Guisasola@uab.cat, JuanAntonio.Baeza@uab.cat

Corresponding author:

Juan A. Baeza

Tel: +34 935811587

Email: juanantonio.baeza@uab.cat

Section S1. Bibliometric snapshot of MCDM methods in Web of Science

To contextualize the methodological choice, a simple title-based bibliometric scan was conducted for four widely used MCDM methods: TOPSIS, VIKOR, PROMETHEE, and WASPAS. Data source and query was made using Web of Science (Core Collection). Title search terms were run separately as TOPSIS, VIKOR, PROMETHEE, and WASPAS. The time window was 2000–2025. Counts include all document types returned by the title query (articles, reviews, proceedings papers, etc.). The annual counts plotted in Figure S1 reflect the number of records whose title contains each term. Across the entire period, TOPSIS consistently dominates in annual publication counts, remains the most frequently appearing method term in titles in recent years, including 71% of the total publications. VIKOR, PROMETHEE, and WASPAS show markedly lower—yet steadily growing—usage. This pattern supports the choice of TOPSIS as a reference ranking core for the present study while noting the active use of alternative MCDM methods in the literature.

Limitations and cautions.

- Title searches are specific but not exhaustive; relevant papers that mention a method only in the abstract/body are missed, and spelling/expansion variants (e.g., "Technique for Order Preference by Similarity to Ideal Solution") are not captured.
- A single paper can mention multiple methods in its title and thus contribute to more than one series.
- Annual totals near the present year can be underestimates due to indexing lag.
- Document-type mix (articles vs. proceedings) is not harmonized here; counts are for quick landscape context rather than a formal bibliometric study.

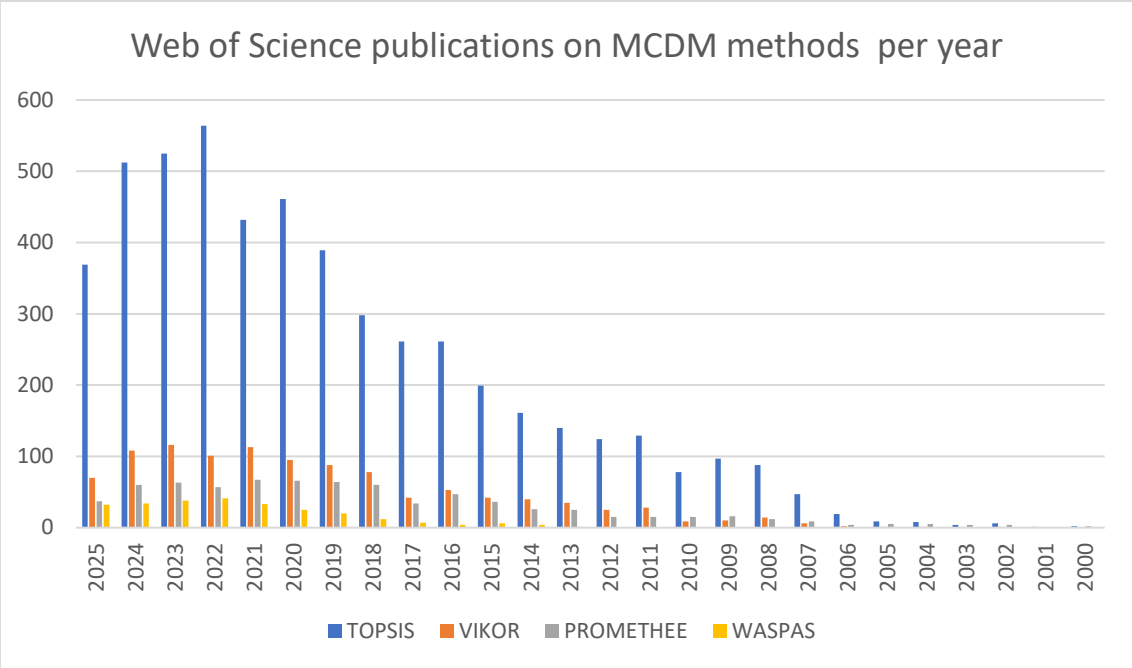

Figure S1. Web of Science title-hits per year (2000–2025) for four MCDM methods (TOPSIS, VIKOR, PROMETHEE, WASPAS). The plot illustrates the sustained popularity of TOPSIS relative to the other methods over the past two decades.

Section S2. Compact mathematical formulation for TOPSIS

Let $C \in \mathbb{R}^{m \times n}$ be the decision matrix, $w \in \mathbb{R}^n$ the weight vector with $w_j \geq 0$ and $\sum_j w_j = 1$, and $MM_j \in \{+1, -1\}$ indicate benefit/cost.

Columns are normalized as $r_{ij} = \frac{MM_j c_{ij}}{\sqrt{\sum_i c_{ij}^2}}$, the weighted matrix is $f_{ij} = w_j r_{ij}$

Positive/negative ideals are $f_j^+ = max_i f_{ij}$ and $f_j^- = min_i f_{ij}$

Distances are $d_i^+ = \sqrt{\sum_j (f_{ij} - f_j^+)^2}$ and $d_i^- = \sqrt{\sum_j (f_{ij} - f_j^-)^2}$

The TOPSIS closeness coefficient is $CC_i = \frac{d_i^-}{d_i^+ + d_i^-}$ and alternatives are ranked by decreasing CC_i.

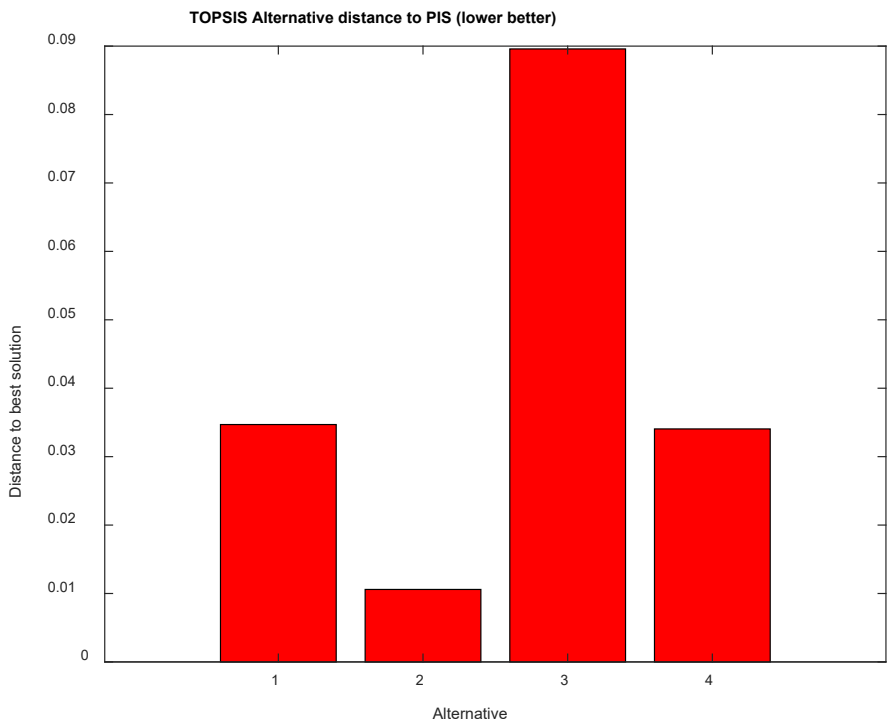

Figure S2. TOPSIS values of the distance to the positive ideal solution (PIS). A lower value indicates that the alternative is closer to the PIS.

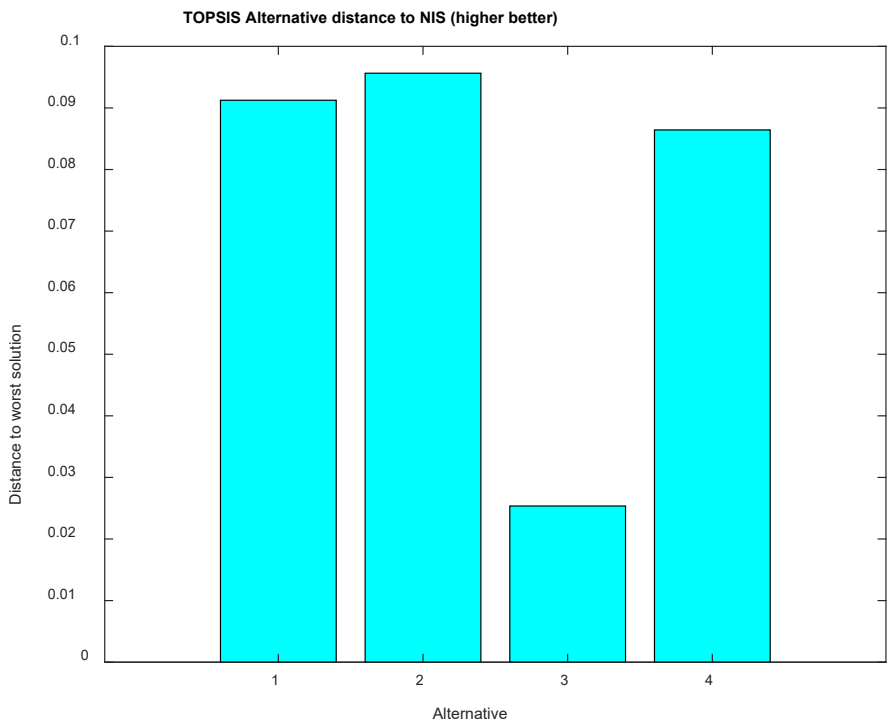

Figure S3. TOPSIS values of the distance to the negative ideal solution (NIS). A higher value indicates that the alternative is far from the NIS and closer to the PIS.

Section S3. TOPSIS ranking for an expert scenario

In this scenario, economic factors are prioritised by using the weights in Table S1. CAPEX and OPEX account for 30% and 20% of the total weight, respectively, rather than 10% each in the equal weight scenario. The nutrient removal percentage (N and P) remains at 10%, while COD removal and the average effluent concentration decrease to 5%, maintaining the total weight at 1. The TOPSIS results in Table S2 show that (2)-A2/O-D remains the best alternative. However, alternative (1)-A2/O-S obtains very close values due to its lower CAPEX, making it more attractive. Closeness coefficient and distance diagnostics for the Table S1 scenario are shown in Figures S4–S6.

Table S1. Expert selection of weights for TOPSIS evaluation

Criterion	*1_CAPEX [k€]*	*2_OPEX [k€]*	*3_COD_{REM} [%]*	*4_N_{REM} [%]*	*5_P_{REM} [%]*	*6_COD_{AVG} [mg/L]*	*7_$N_{T\text{-}AVG}$ [mg/L]*	*8_$P_{T\text{-}AVG}$ [mg/L]*	*9_P_{ROB} [kg]*	*10_N_{ROB} [kg]*
Weight	0.30	0.20	0.05	0.10	0.10	0.05	0.05	0.05	0.05	0.05

Table S2. Results and ranking of alternatives obtained with TOPSIS using the expert weight selection

Alternative	**Description**	**Closeness coefficient**	**Distance to PIS d_i^+**	**Distance to NIS d_i^-**	**Ranking**
(1)-A2/O-S	Anaerobic/anoxic/oxic (A2/O) configuration with a single anaerobic reactor	0.9055	0.0181	0.1732	2
(2)-A2/O-D	Anaerobic/anoxic/oxic (A2/O) configuration with two anaerobic reactors	0.9393	0.0109	0.1695	1
(3)-BARD	Bardenpho configuration	0.2698	0.1533	0.0566	4
(4)-UCT	University of Cape Town configuration	0.6165	0.0886	0.1425	3

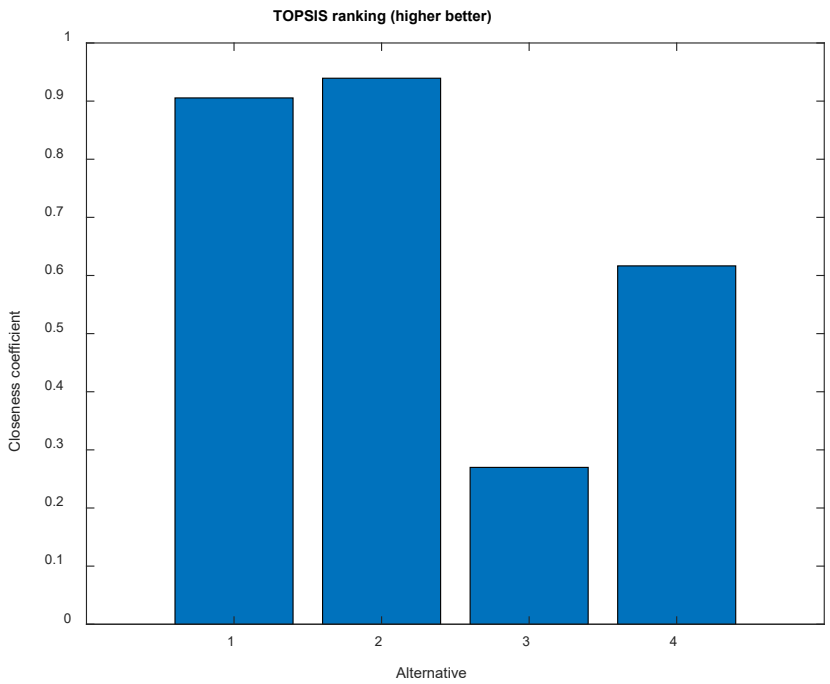

Figure S4. TOPSIS Closeness Coefficients (CC) for the four alternatives studied using the combination of weights of Table S1. Alternative (2)-A2/O-D is still the best option, but alternative (1)-A2/O-S is very close.

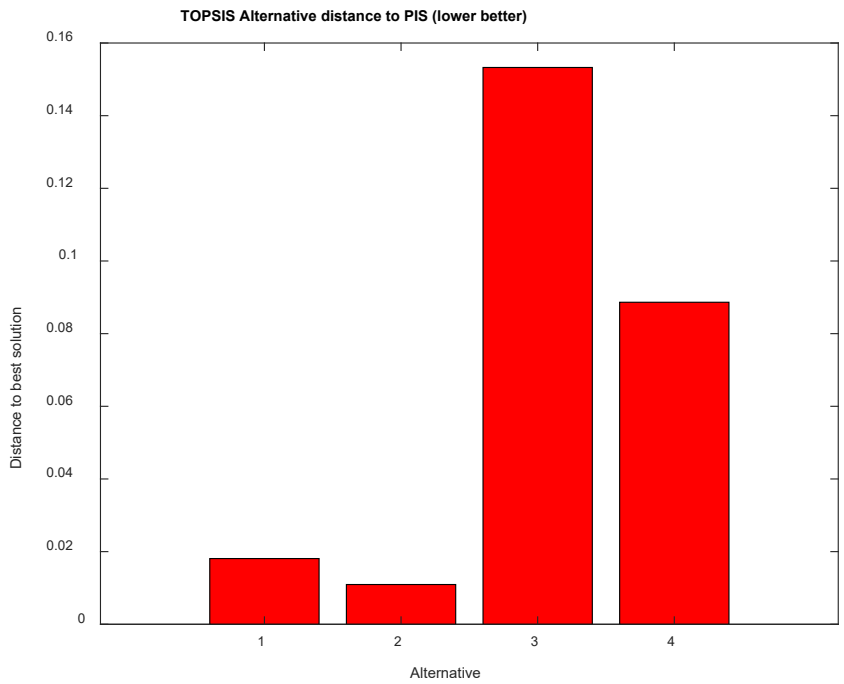

Figure S5. TOPSIS values of the distance to the positive ideal solution (PIS) for the weight scenario of Table S1. A lower value indicates that the alternative is closer to the PIS.

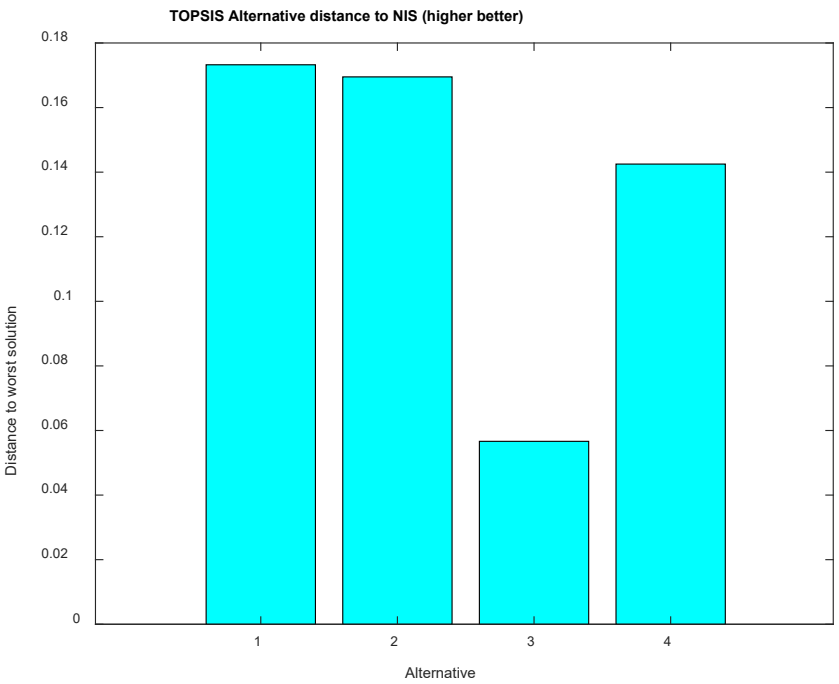

Figure S6. TOPSIS values of the distance to the negative ideal solution (NIS) for the weight scenario of Table S1. A higher value indicates that the alternative is far from the NIS and closer to the PIS.

Table S3. Results of the WSA sensitivity analysis. Default represents the initial weights used to find the best alternative ((2)A2/O-D), Max. weight the highest and Min. weight the lowest values for each criterion for (2)A2/O-D to remain as the best choice.

	CAPEX ***(w_1)***	***OPEX*** ***(w_2)***	***COD_{REM}*** ***(w_3)***	***N_{REM}*** ***(w_4)***	***P_{REM}*** ***(w_5)***	***COD_{AVG}*** ***(w_6)***	***$N_{T\text{-}AVG}$*** ***(w_7)***	***$P_{T\text{-}AVG}$*** ***(w_8)***	***P_{ROB}*** ***(w_9)***	***N_{ROB}*** ***(w_{10})***
Max. weight	0.52	1.00	0.98	1.00	0.55	0.93	1.00	1.00	0.27	1.00
Default	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
Min. weight	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

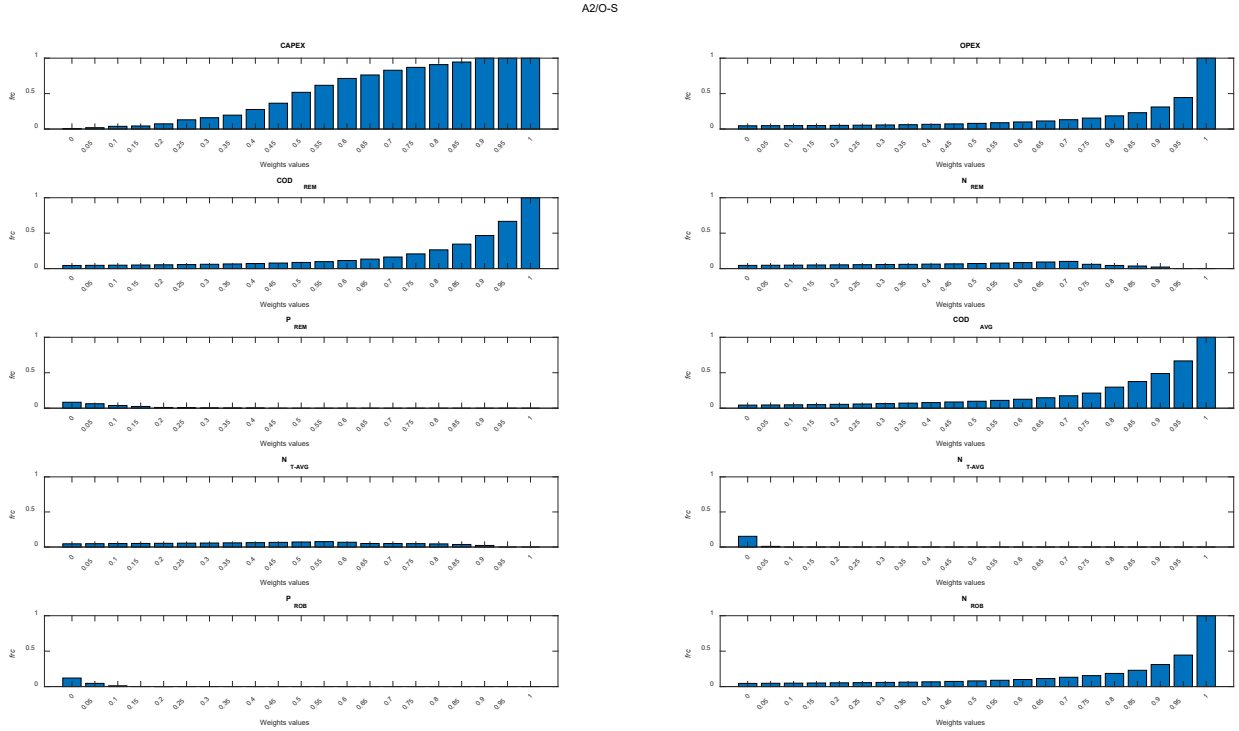

Figure S7. FRWSA-TOPSIS dominance frequency (f_{RC}) results for alternative (1)-A2/O-S.

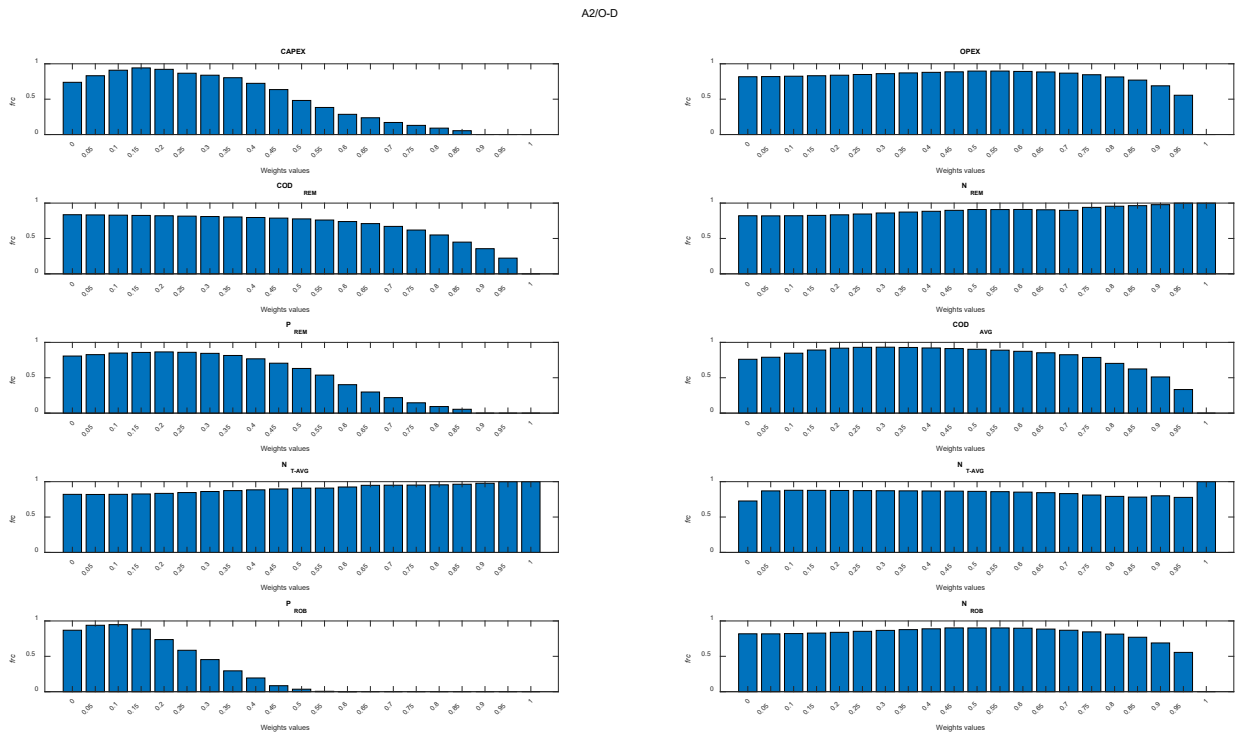

Figure S8. FRWSA-TOPSIS dominance frequency (f_{RC}) results for alternative (2)-A2/O-D.

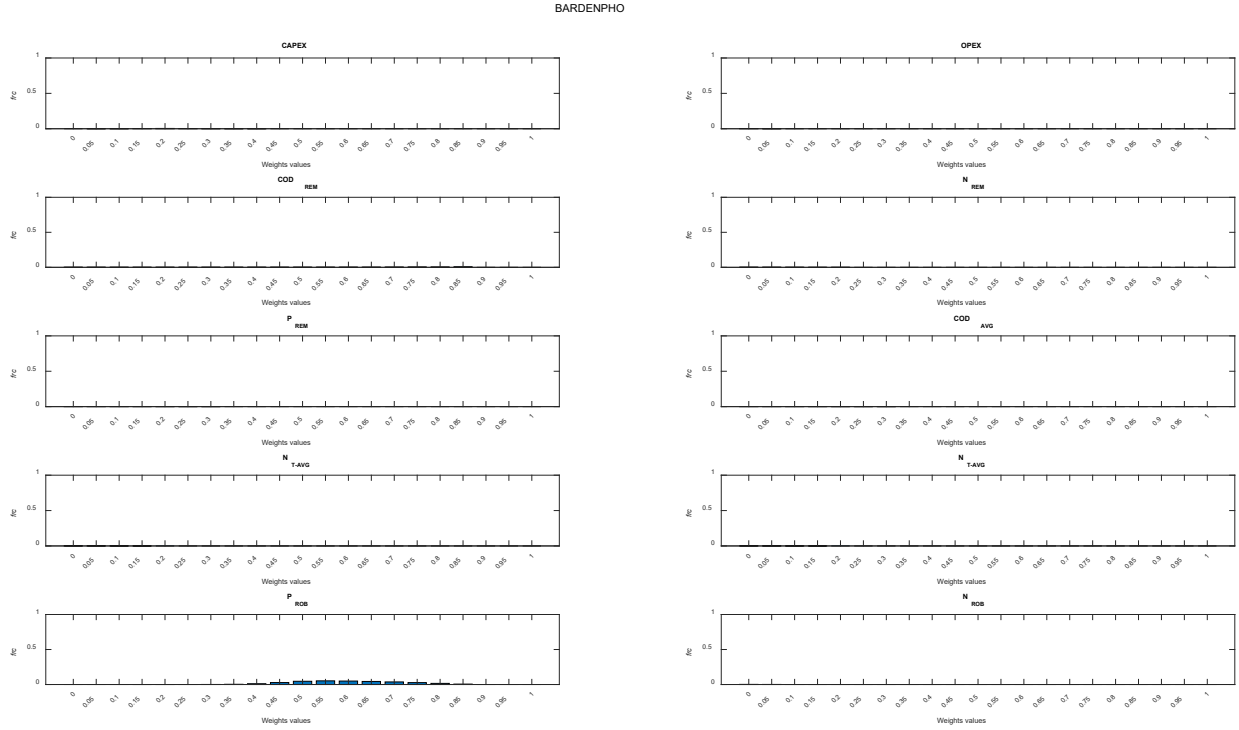

Figure S9. FRWSA-TOPSIS dominance frequency (f_{RC}) results for alternative (3)-BARD.

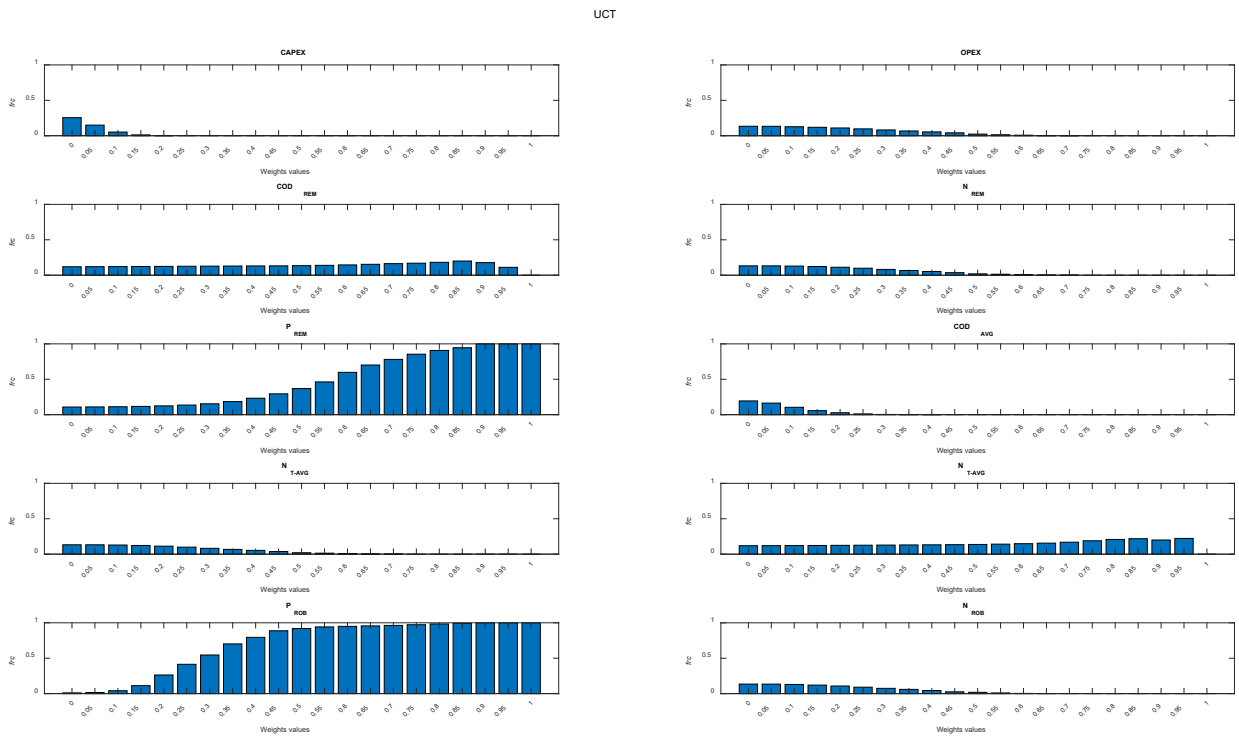

Figure S10. FRWSA-TOPSIS dominance frequency (f_{RC}) results for alternative (4)-UCT.

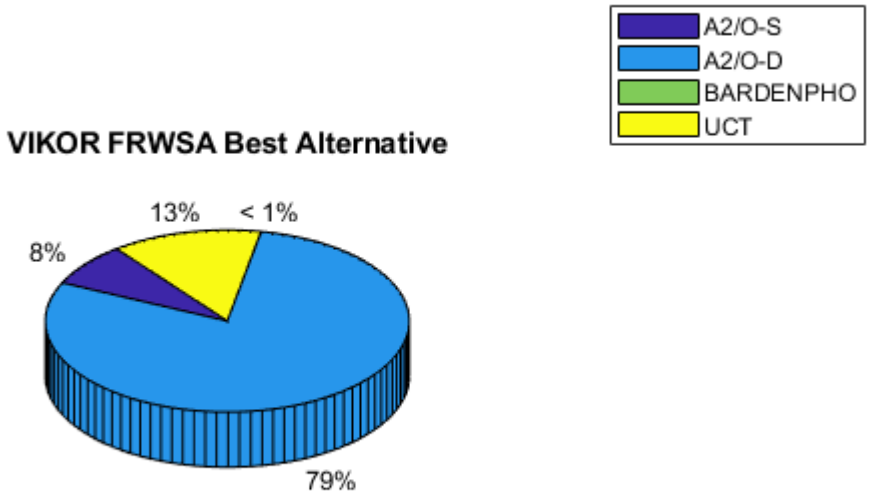

Figure S11. Percentage distribution of the first position of each alternative obtained with the FRWSA-VIKOR method

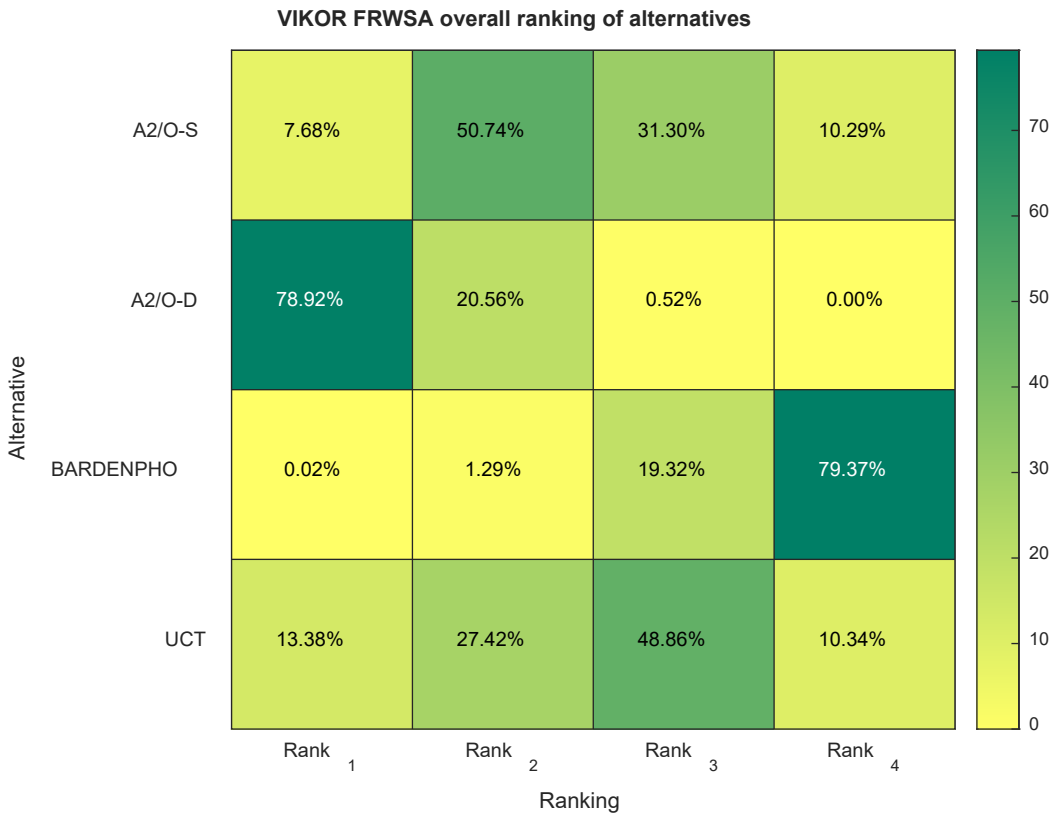

Figure S12. Frequency distribution of the ranking position for each alternative with the FRWSA-VIKOR method

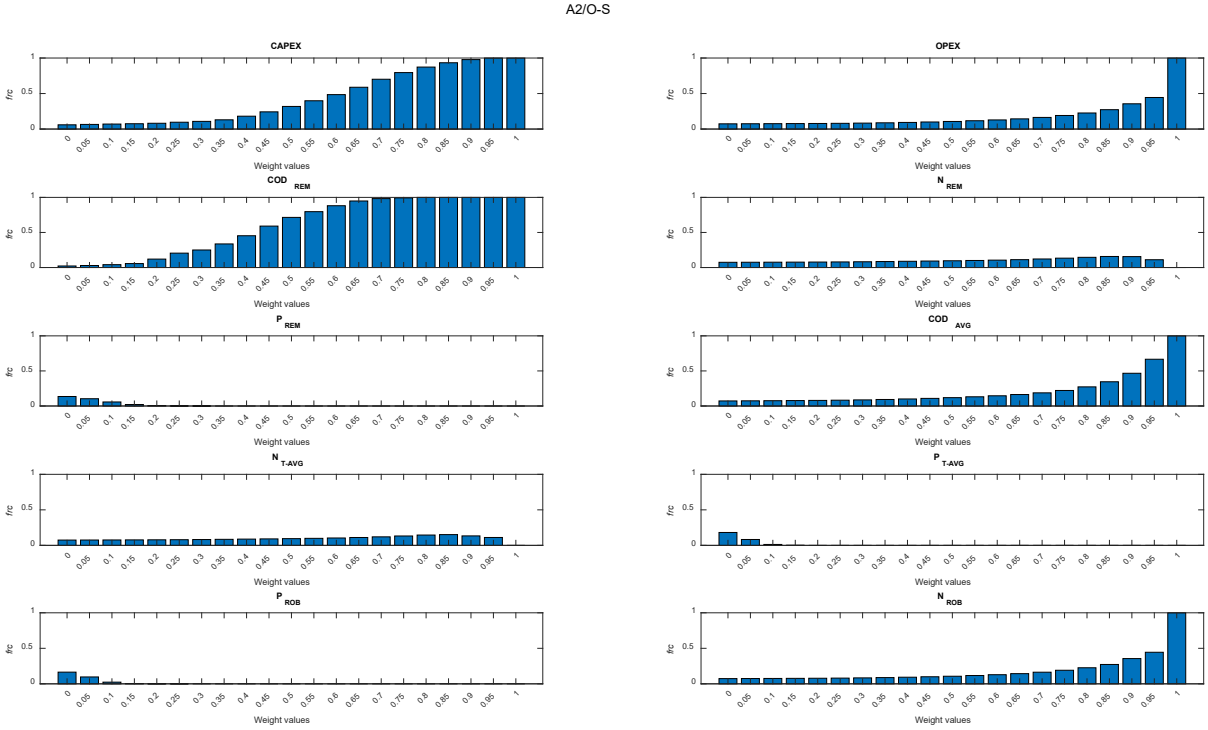

Figure S13. FRWSA-VIKOR dominance frequency (f_{RC}) results for alternative (1)-A2/O-S.

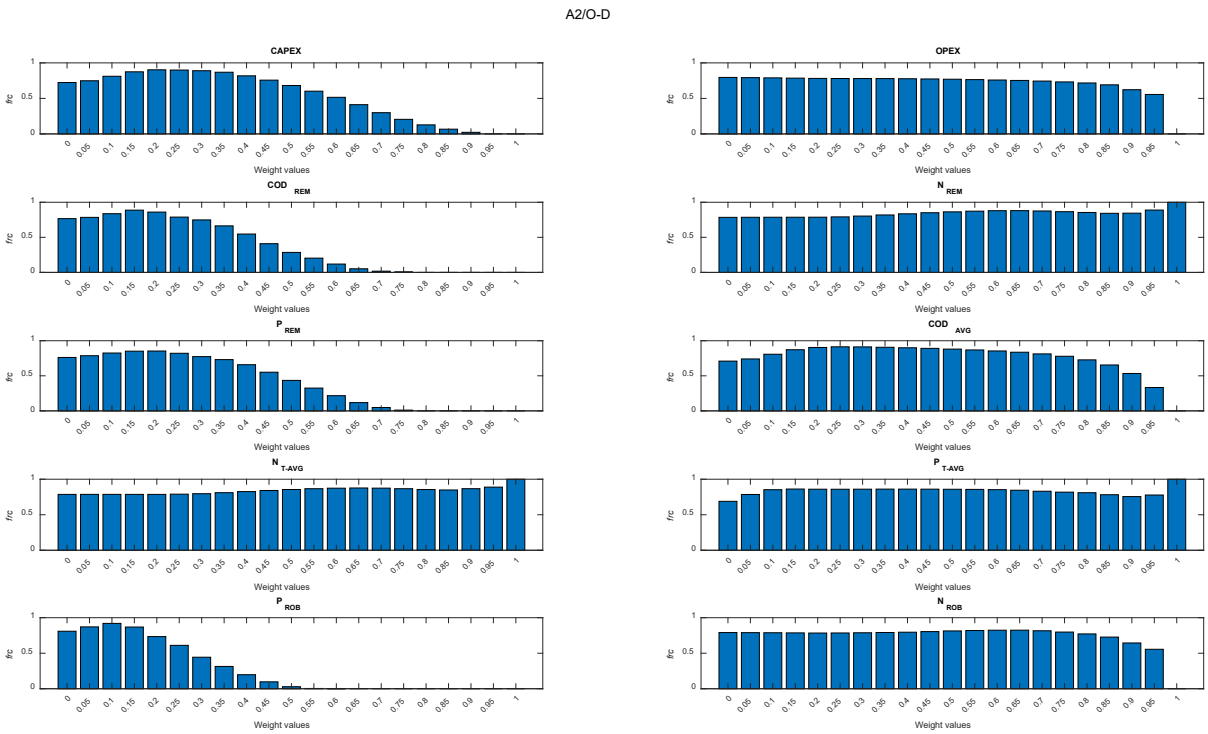

Figure S14. FRWSA-VIKOR dominance frequency (f_{RC}) results for alternative (2)-A2/O-D.

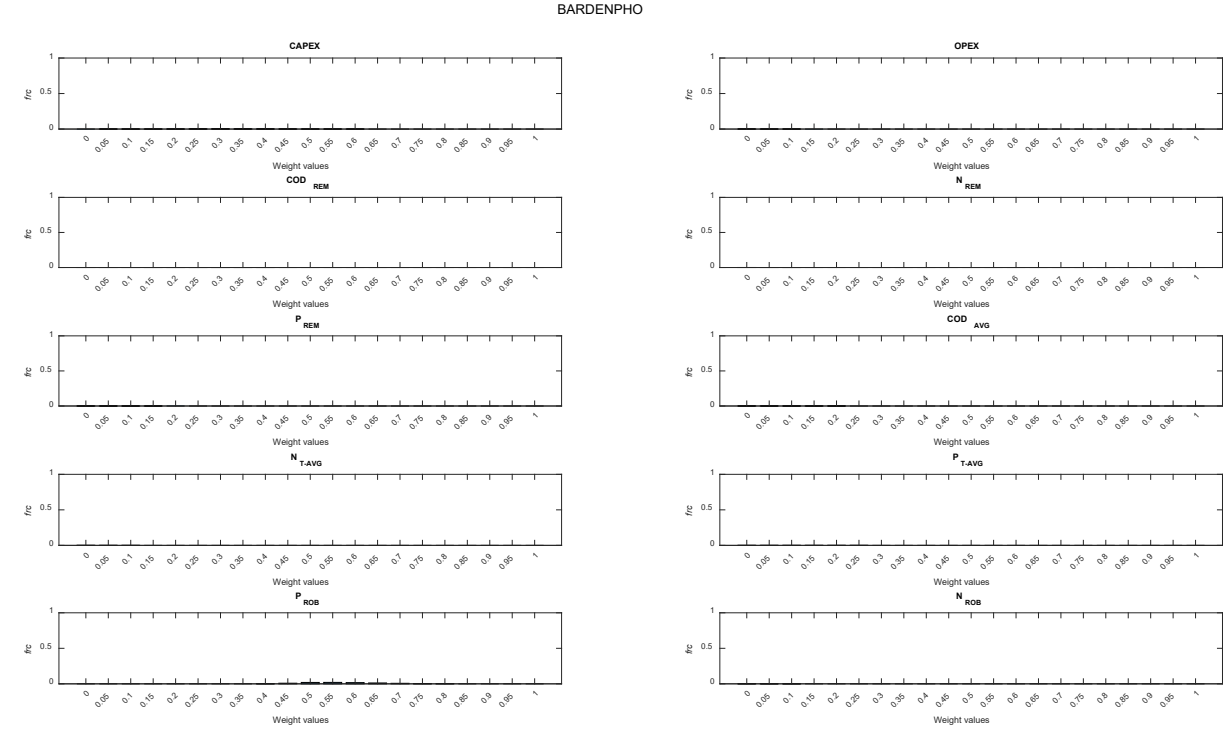

Figure S15. FRWSA-VIKOR dominance frequency (f_{RC}) results for alternative (3)-BARD.

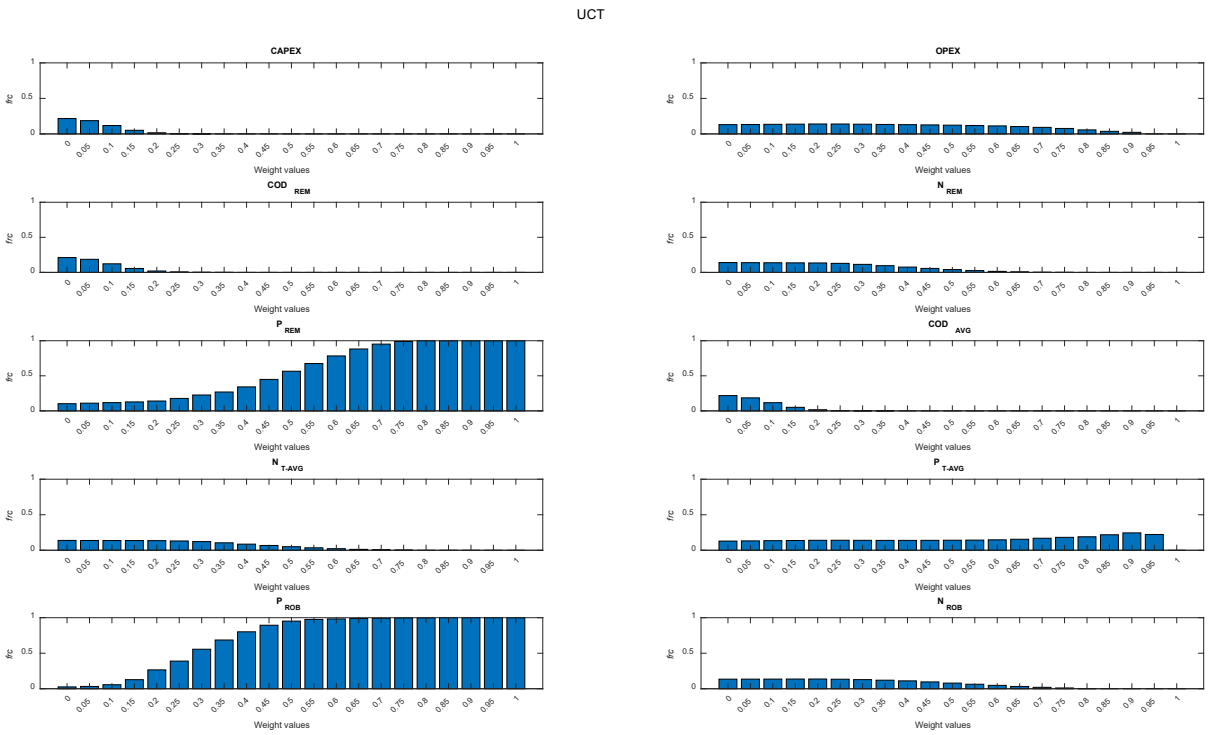

Figure S16. FRWSA-VIKOR dominance frequency (f_{RC}) results for alternative (4)-UCT.

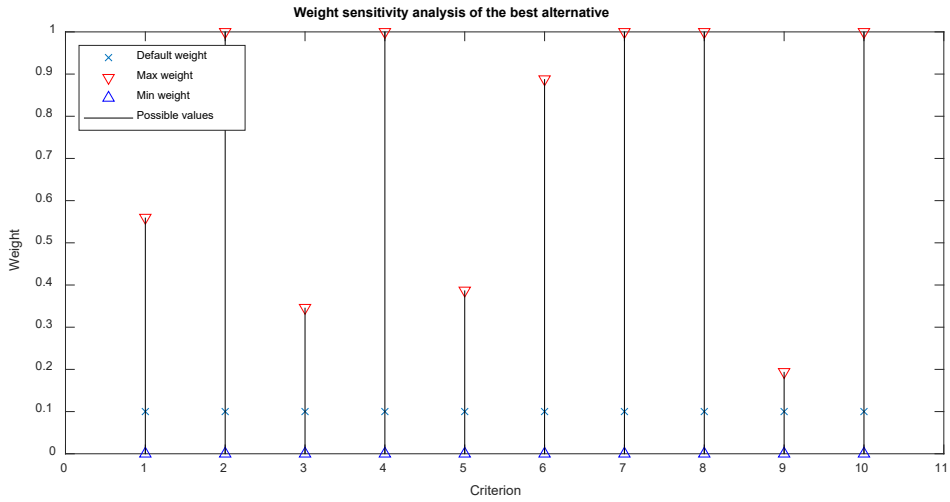

Figure S17. VIKOR Weight sensitivity analysis (WSA) results for the selection of WWTP alternatives. Default weight represents the original weights used that resulted in the selection of (2)-A2/O-D as the best alternative. The line indicates all the weight values obtained following the procedure detailed in Section 3.1 that maintain (2)-A2/O-D as the best choice, with Max and Min weight as the limiting values for these ranges.